



SOCI 350: Statistics in Social Research

Fall 2025

Classroom: STBIO S3/3

Time: Wednesdays and Fridays 2:30 PM – 4:00 PM

Professor: Orsola Torrisi

Email: orsola.torrisi@mcgill.ca

I typically respond to emails within 24 hours (weekdays). If you have not heard from me after 24 hours, you may email me again to make sure I received your message.

Student hours (SH): Wednesdays, 5-6:15 PM

Office: Room 814, LEA

Appointment: Please schedule an appointment at this [link](#). Note: Appointments are 15 minutes long. If you require more time, feel free to book two consecutive slots. Please indicate whether you prefer to meet in person or via Zoom, and briefly mention the topic you would like to discuss.

Teaching Assistants (TAs):

Zekai Lu – SH: Mondays, 3-4 PM, 111A LEA, email: zekai.lu@mail.mcgill.ca

Yi-Cheng (Ken) Hsieh – SH: Fridays, 1-2 PM, 112C LEA, email: yi-cheng.hsieh@mcgill.ca

1 Course Description

This course introduces students to the basics of statistics and probability, which are building blocks for much of the research that is done in the Social Sciences. The course is divided into two broad parts that give students an overview of (i) descriptive and (ii) inferential statistics, with emphasis on conceptual understanding and statistical thinking. Real data from several social science disciplines will be integrated and studied throughout the course using state-of-the-art software tools. At the end of the semester students should know the difference between populations and samples, how to design experiments, how to summarise and analyse the results of a statistical experiment, how to test basic hypotheses and how to communicate statistical insights.

1.1 Who is this course for?

- Students who want to be able to better read and interpret quantitative research in the social sciences (even if you only intend to do qualitative work yourself).
 - Students who are willing to spend time outside the classroom to learn the course materials, as data analysis is a skill learned through actual practice.
 - Students who want to be able to apply quantitative methods to their papers and future careers.
 - Students who have never taken a semester-long course on quantitative methods for data analysis.
- If you have, you are advised not to enroll.

1.2 Prerequisites

This is an introductory course, so no previous experience in statistics is required, but we will assume some familiarity with elementary algebraic notation. The course serves as a background for further statistics and data-analysis courses, helping to provide the intuition that can sometimes be lost amid the formulas.

2 Course Goals

The course objectives include:

- Learn the basic tools of empirical research in the social sciences. Specifically, students will acquire the ability to distinguish between populations and samples, develop skills in experimental design, learn how to summarise and analyse statistical findings, gain proficiency in fundamental hypotheses testing, and effectively communicate statistical insights.
- Gain competency in the critical evaluation of empirical analyses of the social world you live in.
- Obtain the tools—both technical and intellectual—to perform statistical analyses and critiques of your own.
- Gain real-world skills (such as coding or project-management skills) that will help you obtain jobs in careers of the future.

Overall, by the end of this course, I hope all students—even those who think they are least excited about studying statistics—will realise that statistics is an applied field with a wide range of practical uses. You do not have to be a math guru to learn from interesting, real data, but you need to be aware that data are often messy and statistical tools are far from perfect. However, when you understand the strengths and weaknesses of these tools, you can use them to learn interesting things about the world and debunk many non-sensical things you that are often presented as facts.

3 Instructional Method

The course will be organised as a mix of (i) lectures in which I present the conceptual tools behind specific sets of statistical analyses, and (ii) applied demonstration/exercise/lab sessions aimed at helping you learn how to perform these analyses using statistical software (specifically **Stata**).

The class tutorial will be mostly delivered by me or your TAs during class time. **We aim to have a lab session after every 3 lectures. The tasks and analyses performed in Stata during the labs will be similar to those of the individual assignments.** After labs, I will post exercises (non-graded) that will help you test and solidify your knowledge. I highly recommend you do them as some commands will appear again in your assignments. You will find the demonstration/exercise sessions more useful if you can follow on your computers and be active participants in the exercises provided. Therefore, I highly encourage you to have your computers (no tablets as the software does not work there) with you for the applied/lab sessions.

3.1 Textbook

All the material you will need to know will be detailed in my slides and will be discussed in class. However, I will draw from two main textbook resources:

- Moore, D., McCabe, G., and Craig, B.A. (2021). *Introduction to the Practice of Statistics*, 10th edition. WH Freeman: New York. – This is the book I will follow a bit more closely for the lectures. The main e-version of the book will be available on MyCourses. Older versions of Moore et al. are available in McGill's library ([here](#)). Alternatively, you can purchase copies of the 10th edition on Amazon ([here](#)) or on the publisher's website ([here](#)).
- Healey, J., Donoghue, C. and Prus, S. (2022). *Statistics A Tool for Social Research and Data Analysis*, 5th Canadian edition. Toronto: Nelson Education. – A slightly easier book. You can access the online version of the book [here](#). Hard copies of the 5th edition can be purchased on Amazon ([here](#)) or on the publisher's website ([here](#)).

If you find cheaper but older editions of either book, that is also fine, but it your responsibility to be aware of any discrepancies.

Optional resources:

- Acock, A. C. (2008). *A gentle introduction to Stata*. Stata press.
(PDF available [here](#) or on MyCourses.)
- Pevalin, D. and Robson, K. (2009). *The Stata Survival Manual*. McGraw–Hill.

These two resources provide guidance on how to use **Stata**, a statistical software widely used in the social sciences. Note: These books are optional, as both myself and the TAs will guide you towards learning **Stata** both in class and during office hours. Also, there are multiple online resources that are good substitutes of this book—feel free to ask us if interested.

All other resources used, including lecture notes, practical exercises, datasets, homework assignments, and additional references, will be posted on MyCourses. As for the lecture notes, I will attempt to post them the day before each class, so you can take notes on them. I will post revised overheads after the class if any change or correction is to be made. Please note that it is your responsibility to familiarise yourself with MyCourses.

3.2 Preparing for lectures

One way to prepare for class is to read the chapters ahead of time. After class, read the chapter again, test yourself with the weekly practice questions and do any suggested practice exercises; do additional problems if you are having trouble. Then read the text of the chapter again to solidify what you learned. At the beginning of every class, I will do my best to review briefly what we have covered in the previous class. I will also do my best to upload slides ahead of class.

3.3 Software

The software we will use in this class for demonstration/exercise sessions and that I also expect you to utilise to complete assignments is called **Stata**. This software is available for all McGill students to access remotely. The Faculty of Arts has purchased a license that covers all students. Instructions to install and apply McGill's license are [here](#) as well as on MyCourses.

You can also purchase a temporary student license to use **Stata** on your own computers, however, this should not be needed with the remote access provided by McGill. If you still wish to purchase **Stata** for

this class, **Stata**/IC will work. The price for a temporary Stata/IC license is CAD 48 for six months (see [here](#)). If you think you might use **Stata** again in the future, however, I recommend opting for Stata/SE instead of Stata/IC.

3.4 Calculators

You will need a scientific calculator for some assignments. You may want to choose a calculator that also has some statistical functions built in. You are responsible for learning how to use the statistical functions on your calculator. We do not provide specific instructions on how to use calculators.

4 Assignments and Grading

Your success in this class is important to me. You learn from, me, your peers and I learn from you! Hence, assignments for this course are thought to incentivise continuous engagement with the material, regular feedback and peer collaboration. Note that for this course there will be **no final exam**.

Your final grade will be based on several components described below. Please note that the below course requirements are non-negotiable. If you know you cannot meet them, do not take this course

1) Topic-specific practice questions – 10% of final grade

Practice questions will be accessed on MyCourses and will be made available at the end of the first class for Topics 1-5, but due at its end (mostly in conjunction with Lab sessions) as per calendar below.

2) Two “mid-term” quizzes – 15% each, total 30% of final grade

These will be about 40 questions, multiple choice quizzes. We will hold them during class time and will last approximately 1.5 hours. They will be non-cumulative. Exam dates are on the schedule below, but dates may change. There will be **no make-up quizzes**. Quizzes can only be skipped only due to emergency/medical reasons (provided proper justification and/or documentation). In this case, an alternative quiz date will be set up and will take place under the supervision of either me or one of your TAs. A cheat sheet with formulas might be provided for your use during the quizzes; you may use this and a calculator, but no other material. These quizzes may touch on interpreting output from **Stata** but will not test you on its commands.

3) Individual assignments – 15% each, total 30% of final grade

There will be 2 individual assignments (each worth 15%) throughout the semester. I recommend starting working on the assignments as soon as they become available (15 working days before the due date), which will allow you to follow the course as topics are being discussed, and also give you ample time to troubleshoot any possible issues with me and/or the TAs. I will post assignments on MyCourses the day announced and briefly discuss them in class upon request. Unless you meet an exception criterion, no late assignments are accepted.

In both assignments, you will be asked to perform sets of statistical analyses using **Stata**. You can discuss this assignment with other students in the class to further your understanding of the material, but you must write it up independently. To simply copy another student's assignment and turn it in is cheating

(please refer to Academic Integrity Policy below for more details). You will be expected to turn in both your answers to the assignment's questions and the **Stata** syntax you have employed to reach them.

4) Group project – 30% of final grade

One of the goals of this course is to help you develop your skills as a producer of social research. As such, this assignment consists of a data analysis project focused on descriptive statistics and graphical display of data, accompanied by a 4-page write-up (references do not count in the page count) – basically a short academic paper on a research question of your own interest. So, this is the cool part of the course :). Ideally, you will work in teams of 2-5, and each of you will obtain the same grade. Only one write-up and code per group will be submitted.

To simplify things, we will provide you with a series of clean datasets that you can use for your project on a topic of your own interest. Given the range of backgrounds of students in this class, you are also welcome to use your own dataset, but you must share it with us beforehand (no later than Oct 31st!) so we can approve it and provide proper guidance. Further details on this group project will be specified in class during the course. This group project will mark the completion of this course, as there will be no final exam. We will have specific labs to guide you with data, issues that arise when analysing data, cleaning strategies, and analytical strategies.

To summarise all of the above, the grading scheme and dates for homework submission is below:

Component	Weight	Due Date
Topic practice questions	10%	As per calendar below (always due 10 AM on given day)
Mid-term quiz 1	15%	26 September (in class)
Individual assignment 1	15%	Due 22 October @ 10 AM
Mid-term quiz 2	15%	19 November (in class)
Individual assignment 2	15%	Due 26 November @ 10 AM
Group project	30%	Due 12 December @ 10 AM

4.1 Re-grading

Students who wish to contest a grade must do so in writing (by email, sent to me) providing the reasoning behind their challenge to the grade received within two weeks (15 working days) from the day on which the grades are returned (included). The TA who graded the assignment may at that point re-grade your assignment and raise or lower the grade. If you are still unsatisfied after the re-assessment, you can re-submit the assignment to me (original copy with TA comments), along with your justification. I will then re-evaluate the paper but also reserve the right to raise or lower the grade.

4.2 Re-quiz/assignment make-up

Sometimes our lives interfere with our coursework. This is a normal, expected thing that happens for some students every semester. If you find yourself struggling to complete readings or assignments, please come to office hours so we can discuss academic accommodation options that can help you make the most out of the semester. However, please note that without prior discussions, proven emergency and/or medical circumstances, it will not be possible for us to have you make up in-class quizzes. If you are absent for such reasons, I will try my best to arrange an alternative quiz date provided the absence

is properly justified and, when necessary, documented. The alternative quiz will take place under the supervision of either me or one of your TAs.

⇒ If you know you will miss either quizzes or assignments **do not take this course**. No late assignments will be accepted without prior discussion with me and, if needed, proper documentation.

5 Course Policies

5.1 Attendance

If you want to do well in this class and understand the material, it is essential that you come to class. You will notice that the pace will be quite fast, and we will cover a good amount of material. It is however your responsibility to attend any lectures or conferences. As such, attendance will not be recorded, and no clickers will be used. Attendance will not count towards your final grade.

5.2 Communication Policy

I will be available to answer student emails, but I strongly encourage students to use My-Courses to ask questions, so the TAs and I are not asked the same question multiple times. I will set up a discussion board on MyCourses as another means of communication. I encourage you to use this board to ask any questions you may have. With that said, I will do my best to reply to your email within 24 hours. If you have not heard from me after 24 hours (excluding weekends) you may email me again to make sure I received your message. I will also be available by appointment, if students wish to discuss things one on one, outside of office hour times. For questions regarding using **Stata**, it would be best if you contact the TAs who will also have office hours to help you and answer your questions

5.3 Use of Generative AI

One of the primary learning objectives for this course is to help you understand how to code in **Stata**, improve your ability to write clear and accurate summaries of your research findings. In pursuit of these goals, you are permitted to use generative AI, such as ChatGPT, to assist with your research, but you are not permitted to use it to write your essays for you. I also highly discourage you from seeking coding assistance from ChatGPT or similar. They rarely get it right.

You may use generative AI to help you:

- Brainstorm for ideas for your group project if you need
- Get clarification or elaboration on key concepts and ideas
- Find additional references
- Improve your writing by getting feedback on your use of vocabulary, making more coherent arguments, and reducing redundancies.

You may **NOT** use generative AI to:

- Write part or all of your research summaries. All parts of the summary, including your code, must originate from you. Copying AI generated text and claiming it as your own is plagiarism and will be treated as such according to the Academic Integrity Policy provided below.

Caveats on use of generative AI:

- Hallucinations and false information. Generative AI is well-known for confidently asserting made up information. You are responsible for the accuracy of your statements.

Implications for grading and assessment:

- You are required to add a statement at the end of each essay stating whether you used generative AI and, if so, for which permissible activities listed above. This declaration will not influence your grade.
- Hallmarks of good research essay writing (and weaknesses of generative AI) include accuracy, originality, detailed and substantive (not superficial) arguments, clearly stated positions and opinions (not wishy-washy neutral statements), and appropriate integration of references. Hence, all these attributes will be taken into consideration in your grade.

5.4 Extraordinary Circumstances

In the event of extraordinary circumstances beyond the control of McGill University, assessment tasks in a course are subject to change, provided students are sent adequate and timely communications about the change.

5.5 Copyright

All slides, video recordings, lecture notes, etc. remain the instructor's intellectual property. As such, you may use these only for your own learning (and research, with proper referencing/citation) ends. You are not permitted to disseminate or share these materials; doing so may violate the instructor's intellectual property rights and could be cause for disciplinary action.

Instructor generated course materials (e.g., handouts, notes, summaries, exam questions, recorded lectures, etc.) are protected by law and may not be copied or distributed in any form or in any medium without explicit permission of the instructor. Note that infringements of copyright can be subject to follow up by the University under the Code of Student Conduct and Disciplinary Procedures

5.6 Inclusive Learning Environment

As the instructor of this course, I endeavour to provide an inclusive learning environment. However, if you experience barriers to learning in this course, do not hesitate to discuss them with me and the Office for Students with Disabilities, 514-398-6009.

5.7 Language of Submission

In accord with McGill University's Charter of Students' Rights, students in this course have the right to submit in English or in French any written work that is to be graded.

Conformément à la Charte des droits de l'étudiant de l'Université McGill, chaque étudiant a le droit de soumettre en français ou en anglais tout travail écrit devant être noté (sauf dans le cas des cours dont l'un des objets est la maîtrise d'une langue). Les étudiants de ce cours peuvent rédiger tous leurs travaux (incluant les examens) en français, mais doivent pour ce faire obtenir la permission préalable de la professeure. Aucune permission rétroactive ne sera accordée. Indigenous Land Statement

McGill University is located on land which has long served as a site of meeting and exchange amongst Indigenous peoples, including the Haudenosaunee and Anishinaabe nations. McGill honours, recognises and respects these nations as the traditional stewards of the lands and waters on which we meet today.

5.8 Academic Integrity Policy

Just like writing a paper, copying other people's computer code constitutes plagiarism. Moreover, **data cleaning, coding and analysis are learned through trial and error**. Please do not under any circumstance copy another student's code. If you are found to have done so, you may be referred to the appropriate Dean. The instructors reserve the right to use software to compare the code that has been written by different students. Note that to support academic integrity, your assignments may be submitted to text-matching or other appropriate software (e.g., formula-, equation-, and graph-matching). McGill University values academic integrity. Therefore, all students must understand the meaning and consequences of cheating, plagiarism and other academic offences under the Code of Student Conduct and Disciplinary Procedures (see [here](#) for more information, approved by Senate on 29 January 2003).

L'université McGill attache une haute importance à l'honnêteté académique. Il incombe par conséquent à tous les étudiants de comprendre ce que l'on entend par tricherie, plagiat et autres infractions académiques, ainsi que les conséquences que peuvent avoir de telles actions, selon le Code de conduite de l'étudiant et des procédures disciplinaires (pour de plus amples renseignements, veuillez consulter le site [ici](#))

6 Class Schedule Shortened Version (Provisional)

The course is divided into 2 main parts and 6 main Topics as described below. The scheduling of topics for the course and associated readings are subject to change. Any of such change will be announced ahead of time in class and through MyCourses. Students are responsible for keeping abreast of any changes made.

Legend: dates for (i) Lab sessions are highlighted in blue; (ii) mid-term quizzes in light blue, (iii) topic-specific practice questions in green and (iii) individual/group assignments in red.

PART 1 – DESCRIPTIVE STATISTICS

Topic 1 – INTRODUCTION TO DATA, DATA ANALYSIS AND DESCRIPTION

- **Aug 27:** Introduction, class syllabus, class materials, and course overview
- **Aug 29:** Viewing data I (types of variables, measurement levels)
- **Sept 3:** Viewing data II (numerical and graphical description, basic concepts, linear transformations)
- **Sept 5:** The Normal distribution and z-scores
- **Sept 10:** Lab session I (bring laptop) + Topic 1 practice questions due

Topic 2 – RELATIONSHIP IN THE DATA

- **Sept 12:** Describing bivariate relationships
- **Sept 17:** Correlation, regression and causation I
- **Sept 19:** Correlation, regression and causation II and Experiments (my slides)
- **Sept 24:** Lab session II (bring laptop) + Topic 2 practice questions due
- **Sept 26:** First mid-term Quiz + Group work instructions live

PART 2 – FOUNDATIONS OF STATISTICAL INFERENCE

Topic 3 – BASIC PROBABILITY THEORY

- **Oct 1:** Basics of probability and recap of the Normal distribution
NOTE: Oct 2. First individual assignment live (10 am)

Topic 4 – SAMPLING DISTRIBUTIONS

- **Oct 3:** Sampling variability, sampling distributions (means) and Central Limit Theorem
- **Oct 8:** Sampling distributions (proportions), bias and noise

Topic 5 – STATISTICAL CONFIDENCE AND HYPOTHESIS TESTING

- **Oct 10:** Lab session III (bring laptop) + Topic 3-4 practice questions due

Oct 15-17 No class, Fall break

- **Oct 22:** Statistical confidence I + **First individual assignment due**
- **Oct 24:** Statistical confidence II (margin of error)
- **Oct 29:** Hypothesis testing I (alpha levels)
- **Oct 31:** Hypothesis testing II (p-values) + **Deadline to form groups for group project**
NOTE: Nov 4. Second individual assignment live (10 am)
- **Nov 5:** Type I - II errors and statistical power
- **Nov 7:** T-student distribution, T-tests
- **Nov 12:** Chi-square test
- **Nov 14:** **Lab session IV (bring laptop)** + **Topic 5 practice questions due**
- **Nov 19:** **Second mid-term Quiz**
- **Nov 21:** Interpreting linear regression output with **Lab session V (bring laptop)**

Topic 6 – DATA SOURCES

- **Nov 26:** Data sources, sampling strategies + **Second individual assignment due**
- **Nov 28:** **Lab session focusing on your group project**
- **Dec 12:** **Group assignment due**

7 Class Schedule Detailed Version (Provisional)

PART 1 – DESCRIPTIVE STATISTICS

Topic 1 – INTRODUCTION TO DATA, DATA ANALYSIS AND DESCRIPTION

I: Introduction to statistics (Aug. 27)

Readings: Moore et al., Chapter 1 and Healey et al. Chapter 1 (Both optional)

- Course structure and goals
- What is statistics? And why should we care?

II: Viewing data (Aug. 29, Sept. 3)

Readings: Moore et al., Chapter 1 (Sec. 1.1-1.3) and Healey et al. Chapters 1-2-3

- What is data?
- Types of variables: categorical and numerical variables
- Measurement levels
- Univariate description I: numerical description for categorical variables
 - Frequencies, percentages, ratios and rates
- Univariate description II: numerical description for continuous variables
 - Measures of centrality, spread and symmetry
- Univariate description III: graphic description
 - Bar graphs, pie charts, histograms and boxplots
- Basic concepts: Population, samples and random samples
 - Population parameters vs sample statistics
 - Transforming variables: Linear transformation

III: The Normal distribution and z-scores (Sept. 5)

Readings: Moore et al., Chapter 1 (Sec. 1.4) and Healey et al. Chapter 4

- Density curves
- Properties of the normal distribution
- Z-scores

 Sept 10 - Lab session I (bring laptop) – familiarising with Stata and univariate description

 Sept 10 – Topic 1 practice questions due by 10 AM

Topic 2 – RELATIONSHIP IN THE DATA

I: Describing bivariate relationships (Sep. 12)

Readings: Moore et al., Chapter 2 (Sec. 2.1-2.2, 2.6) and Healey et al. Chapter 13.1, 13.2, 8.1, 8.2

***note:** my slides here follow more closely Moore et al., read Healey et al. only after class

- Scatterplots
- Two-way tables

II: Correlations, regressions and causation (Sep. 17-19)

Readings: Moore et al., Chapter 2 (Sec. 2.3-2.4, 2.6-27) and Healey et al. Chapter 13

***note:** my slides here follow more closely Moore et al., read Healey et al. only after class

- Measuring relationships
 - Covariance and correlation
- Introducing linear regression
- Spurious correlation, correlation and causation
 - Omitted variable bias
 - Simpson's paradox

III: Experiments (Sep. 19)

Readings: Moore et al., Chapter 3 (Sec. 3.2), paper on Parachute use to prevent death

- Research question, pros and cons, best practices and ethics
- Design: treatment, controls, randomisation, etc.

⚠ Sept 24 – Lab session II (bring laptop) – univariate description II, graphs

⚠ Sept 24 – Topic 2 practice questions due by 10 AM

⚠ Sept 26 – FIRST MID-TERM QUIZ (in class) + Group work instructions live

PART 2 – FOUNDATIONS OF STATISTICAL INFERENCE**Topic 3 – BASIC PROBABILITY THEORY****I: Introduction to probability theory** (Oct 1)

Readings: Moore et al., Chapter 4 (Sec. 4.1-4.4)

- Basic concepts
 - Vocabulary
 - Probability rules
 - Independence
 - Assigning probabilities (Equally likely events)
- Random variables and probability distributions
- The Normal distribution as a probability distribution

⚠ Oct 2 – First individual assignment goes live (10 AM)

Topic 4 – SAMPLING DISTRIBUTIONS**I: Sampling distribution** (Oct. 3-8)

Readings: Moore et al., Chapter 5 and Healey et al. Chapter 5, Chapter 6.2

- Introduction to inferential statistics
- Sampling variability
- Sampling distributions
- Central limit theorem
- Bias and noise

⚠ Oct 10 – Lab session III (bring laptop)

⚠ Oct 10 – Topics 3-4 practice questions due by 10 AM

Oct 15-17 No class, Fall break

Topic 5 – STATISTICAL CONFIDENCE AND HYPOTHESIS TESTING

I: Statistical confidence (Oct 22-24)

Readings: Moore et al., Chapter 6 (Sec. 6.1) and Healey et al. Chapter 6

***note:** my slides here follow more closely Moore et al., read Healey et al. at the end of the topic.

- Point estimates
- Confidence intervals and margins of error

⚠ Oct 22 – First individual assignment due by 10 AM

II: Hypothesis testing (Oct 29-Nov.5)

Readings: Moore et al., Chapters 6 (Sec. 6.2-6.4), Chapter 7 and Healey et al. Chapter 7.3-7.4, 10-11

***note:** my slides here follow more closely Moore et al., read Healey et al. at the end of the topic

- Hypothesis testing
 - Elements
 - 5 step model
 - One sample and two-sample cases
- Type I and II errors and power

⚠ Nov 4 – Second individual assignment goes live (10 AM)

III: Introduction to inference for regressions (Nov. 7-21)

Readings: Moore et al., Chapter 7 (Sec. 7.1-7.2), Chapter 9, 10 (Sec. 10.1), Chapter 11 and Healey et al. Chapter 6.5-10.4, 13, 14

- T-student distribution and t-test
- Chi-square test
- Interpreting linear regression output with Lab session V (bring laptop)

⚠ Nov 14 – Lab session IV (bring laptop) – CIs and hypothesis testing

⚠ Nov 14 – Topic 5 practice questions due by 10 AM

⚠ Nov 19 – SECOND MID-TERM QUIZ (in class)

⚠ Nov 21 – Lab session V (bring laptop) – Interpreting single and multiple linear regression

Topic 6 – DATA SOURCES

I. Data sources and sampling design (Nov. 26)

Readings: Moore et al., Chapter 3 (Sec. 3.1, 3.2) and Healey et al. Chapter 5 (Sec. 5.1-5.2), Chapter “Other EPSEM Sampling Techniques: Systematic, Stratified, and Cluster Samples”

- Sampling design:

- Random, stratified and cluster sampling
- Sampling weights and representative statistics
- Issues: under-coverage, biases

 Nov 26 – Second individual assignment due by 10 AM

 Nov 28 – Lab session focused on diagnostics for your group project (in class)

 Dec 12 – Final group assignment due by 10 AM